Property-Based Evaluation of Balance Updates

Aim:

To build a system using COBOL that evaluates account balance transfers against formal property invariants, verifying that total combined balance is conserved and neither account becomes negative.

Algorithm:

Step 1: Start the program.
Step 2: Define or accept the initial values for Source Balance (SRC-BAL), Destination Balance (DST-BAL), and Transfer Amount (IN-AMT).
Step 3: Calculate the pre-transfer total balance (PRE-SUM = SRC-BAL + DST-BAL).
Step 4: Validate the transfer conditions:

- If IN-AMT is greater than 0 AND SRC-BAL is greater than or equal to IN-AMT, subtract IN-AMT from SRC-BAL, add IN-AMT to DST-BAL, and set status to "APPROVED".
- Otherwise, keep balances unchanged and set status to "REJECTED".

Step 5: Evaluate the invariant properties:

- Verify that SRC-BAL + DST-BAL equals PRE-SUM (conservation of balance).
- Verify that SRC-BAL >= 0 AND DST-BAL >= 0 (non-negativity constraint).
- If all invariants hold, set invariant check to "PASS"; otherwise, set it to "FAIL".

Step 6: Display the transaction status, initial and final balances, and the invariant check verdict.
Step 7: Repeat Steps 2 through 6 for all test combinations.
Step 8: Stop the program.

Code:

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IDENTIFICATION DIVISION.
PROGRAM-ID. SHORT-TRANSFER.

DATA DIVISION.
WORKING-STORAGE SECTION.
01  TEST-DATA.
   05 FILLER PIC X(24) VALUE "000800000002000000010000".
   05 FILLER PIC X(24) VALUE "000500000001500000050000".
   05 FILLER PIC X(24) VALUE "000300000004000000035000".
   05 FILLER PIC X(24) VALUE "001000000005000000000000".
01  TEST-TABLE REDEFINES TEST-DATA.
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    05 TC OCCURS 4 TIMES.
       10 IN-SRC     PIC 9(6)V99.
       10 IN-DST     PIC 9(6)V99.
       10 IN-AMT     PIC 9(6)V99.

01  TEST-NUM        PIC 9 VALUE 1.
01  SRC-BAL         PIC S9(6)V99.
01  DST-BAL         PIC S9(6)V99.
01  PRE-SUM         PIC S9(7)V99.

01  STATUS-TXT      PIC X(8).
01  PROP-TXT        PIC X(4).

01  D-IN-SRC        PIC ZZZ,ZZ9.99.
01  D-IN-DST        PIC ZZZ,ZZ9.99.
01  D-IN-AMT        PIC ZZZ,ZZ9.99.
01  D-OUT-SRC       PIC ZZZ,ZZ9.99.
01  D-OUT-DST       PIC ZZZ,ZZ9.99.

PROCEDURE DIVISION.
MAIN-LOGIC.
   PERFORM VARYING TEST-NUM FROM 1 BY 1 UNTIL TEST-NUM > 4
      MOVE IN-SRC(TEST-NUM) TO SRC-BAL
      MOVE IN-DST(TEST-NUM) TO DST-BAL
      COMPUTE PRE-SUM = SRC-BAL + DST-BAL

      IF IN-AMT(TEST-NUM) > 0 AND SRC-BAL >= IN-AMT(TEST-NUM)
         SUBTRACT IN-AMT(TEST-NUM) FROM SRC-BAL
         ADD      IN-AMT(TEST-NUM) TO   DST-BAL
         MOVE "APPROVED" TO STATUS-TXT
      ELSE
         MOVE "REJECTED" TO STATUS-TXT
      END-IF

      IF (SRC-BAL + DST-BAL = PRE-SUM) AND
        (SRC-BAL >= 0) AND (DST-BAL >= 0)
         MOVE "PASS" TO PROP-TXT
      ELSE
         MOVE "FAIL" TO PROP-TXT
      END-IF

      MOVE IN-SRC(TEST-NUM)  TO D-IN-SRC
      MOVE IN-DST(TEST-NUM)  TO D-IN-DST
      MOVE IN-AMT(TEST-NUM)  TO D-IN-AMT
      MOVE SRC-BAL           TO D-OUT-SRC
      MOVE DST-BAL           TO D-OUT-DST
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```
    DISPLAY "TEST " TEST-NUM ": " STATUS-TXT
        " | (" FUNCTION TRIM(D-IN-SRC) ", "
            FUNCTION TRIM(D-IN-DST) ") -["
            FUNCTION TRIM(D-IN-AMT) "]-> ("
            FUNCTION TRIM(D-OUT-SRC) ", "
            FUNCTION TRIM(D-OUT-DST) ")"
        " | Invariant Check: " PROP-TXT
END-PERFORM.

STOP RUN.
```

Output:

```
STDIN
300.00
400.00
350.00

Output:
TEST 1: APPROVED | (800.00, 200.00) -[100.00]-> (700.00, 300.00) | Invariant Check: PASS
TEST 2: APPROVED | (500.00, 150.00) -[500.00]-> (0.00, 650.00) | Invariant Check: PASS
TEST 3: REJECTED | (300.00, 400.00) -[350.00]-> (300.00, 400.00) | Invariant Check: PASS
TEST 4: REJECTED | (1,000.00, 500.00) -[0.00]-> (1,000.00, 500.00) | Invariant Check: PASS
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Result:

The COBOL program for property-based evaluation of balance transfers was successfully developed, compiled, and executed.